SE-Lab-7
Fuzzing for Input Validation Bugs

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1. Run the buggy application .py in the repo assigned to you (under Lab7):

Command: python processor.py

```
(venv) abhay@Abhay-Inspiron-DELLO:~/SE/L6_L7_02_car_rental_rate/Lab 7$
(venv) abhay@Abhay-Inspiron-DELLO:~/SE/L6_L7_02_car_rental_rate/Lab 7$ python3 processor.py
==== Running sanitize_license_plate ====
Enter license plate with delimiters (-, _, #): KA-01-AB-1234
Sanitized Plate: KA01AB1234

==== Running parse_trip_mileage ====
Enter CSV of mileages (e.g. 100.5, 45.2, 80.0): 100.5, 45.2, 80.0
Parsed Mileage List: [100.5, 45.2, 80.0]

==== Running format_rental_invoice ====
Enter rental record (Driver:Days:Category): Anirudh:3:SUV
Rental Invoice: Driver Anirudh booked SUV for 3 days: Total $150.00
(venv) abhay@Abhay-Inspiron-DELLO:~/SE/L6_L7_02_car_rental_rate/Lab 7$
```

2. Run the test_processor.py :

Command: python3 -m pytest test_processor.py

```
(venv) abhay@Abhay-Inspiron-DELLO:~/SE/L6_L7_02_car_rental_rate/Lab 7$ python3 -m pytest test_processor.py
============================================================================= test session starts ==============================================================================
platform linux -- Python 3.14.4, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/abhay/SE/L6_L7_02_car_rental_rate/Lab 7
plugins: platformdirs-4.13.3, hypothesis-6.170.15, cov-7.1.0
collected 3 items

test_processor.py FFF                                                                                                                                                  [100%]

================================================================================= FAILURES =================================================================================
_____________________________________________________________ test_sanitize_license_plate_no_crash _____________________________________________________________

s = None

    @given(st.text() | st.none())
    def test_sanitize_license_plate_no_crash(s):
        #To-Do:Using try and except complete this function.
        try:
>           sanitize_license_plate(s)

test_processor.py:8:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

plate = None

    def sanitize_license_plate(plate):
>       plate = plate.strip()
                ^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

processor.py:2: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_sanitize_license_plate_no_crash(s):
                ^^^

test_processor.py:5:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

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- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

    @given(st.text() | st.none())
    def test_sanitize_license_plate_no_crash(s):
        #To-Do:Using try and except complete this function.
        try:
            sanitize_license_plate(s)
        except Exception as e:
>           assert False, f"sanitize_license_plate crashed with: {e}"
E           AssertionError: sanitize_license_plate crashed with: 'NoneType' object has no attribute 'strip'
E           assert False
E           Falling test case: test_sanitize_license_plate_no_crash(
E               s=None,
E           )

test_processor.py:10: AssertionError
_____________________________________________________________ test_parse_trip_mileage_safe _____________________________________________________________

s = None

    @given(st.text() | st.none())
    def test_parse_trip_mileage_safe(s):
        #To-Do:Using try and except complete this function.
        try:
>           parse_trip_mileage(s)

test_processor.py:16:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

mileage_csv = None

    def parse_trip_mileage(mileage_csv):
>       parts = mileage_csv.split(',')
                ^^^^^^^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'split'

processor.py:8: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_parse_trip_mileage_safe(s):
                ^^^

test_processor.py:13:
```

```
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

s = None

    @given(st.text() | st.none())
    def test_parse_trip_mileage_safe(s):
        #To-Do:Using try and except complete this function.
        try:
            parse_trip_mileage(s)
        except (ValueError, TypeError):
            pass
        except Exception as e:
>           assert False, f"parse_trip_mileage crashed unexpectedly with: {e}"
E           AssertionError: parse_trip_mileage crashed unexpectedly with: 'NoneType' object has no attribute 'split'
E           assert False
E           Falling test case: test_parse_trip_mileage_safe(
E               s=None,
E           )

test_processor.py:20: AssertionError
_____________________________________________________________________ test_format_rental_invoice_safe _____________________________________________________________________

s = None

    @given(st.text() | st.none())
    def test_format_rental_invoice_safe(s):
        #To-Do:Using try and except complete this function.
        try:
>           format_rental_invoice(s)

test_processor.py:26:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
```

```
record = None

    def format_rental_invoice(record):
>       parts = record.split(':')
                ^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'split'

processor.py:12: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_format_rental_invoice_safe(s):
                  ^^^

test_processor.py:23:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

s = None

    @given(st.text() | st.none())
    def test_format_rental_invoice_safe(s):
        #To-Do:Using try and except complete this function.
        try:
            format_rental_invoice(s)
        except (ValueError, TypeError, IndexError):
            pass
        except Exception as e:
>           assert False, f"format_rental_invoice crashed unexpectedly with: {e}"
E           AssertionError: format_rental_invoice crashed unexpectedly with: 'NoneType' object has no attribute 'split'
E           assert False
E           Falling test case: test_format_rental_invoice_safe(
E               s=None,
E           )

test_processor.py:30: AssertionError
=========================================================================== short test summary info ============================================================================
FAILED test_processor.py::test_sanitize_license_plate_no_crash - AssertionError: sanitize_license_plate crashed with: 'NoneType' object has no attribute 'strip'
FAILED test_processor.py::test_parse_trip_mileage_safe - AssertionError: parse_trip_mileage crashed unexpectedly with: 'NoneType' object has no attribute 'split'
FAILED test_processor.py::test_format_rental_invoice_safe - AssertionError: format_rental_invoice crashed unexpectedly with: 'NoneType' object has no attribute 'split'
============================================================================ 3 failed in 0.27s =============================================================================
```

3. Fix the buggy application python script:

Run the fixed processor.py –
Command : python processor.py



4. Re-Run the test_processor.py wrt to fixed processor and take the ss of the output:

Command: python3 -m pytest test_processor.py

